FROM RESEARCH TO INDUSTRY

SARAF-LINAC PROJECT ORAL POSTER (MOPLR075)

LINAC 2016 CONFERENCE

EAST LANSING, MI, USA

laboratoire commun CEA/DSM CNRS/IN2P3

Presented by N. Pichoff

SEPTEMBER 26, 2016

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DELIVRABLES

SARAF Top Level Requirements :

- Beam:
  - Deuterons/protons,
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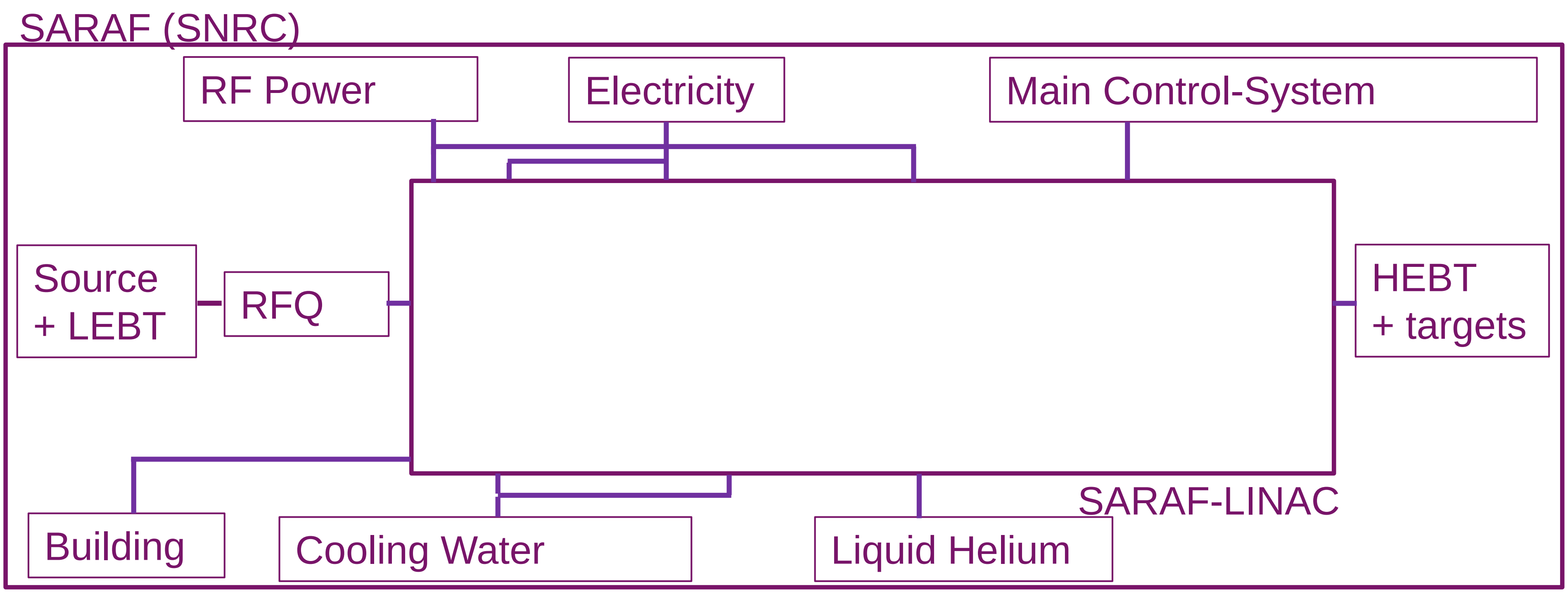

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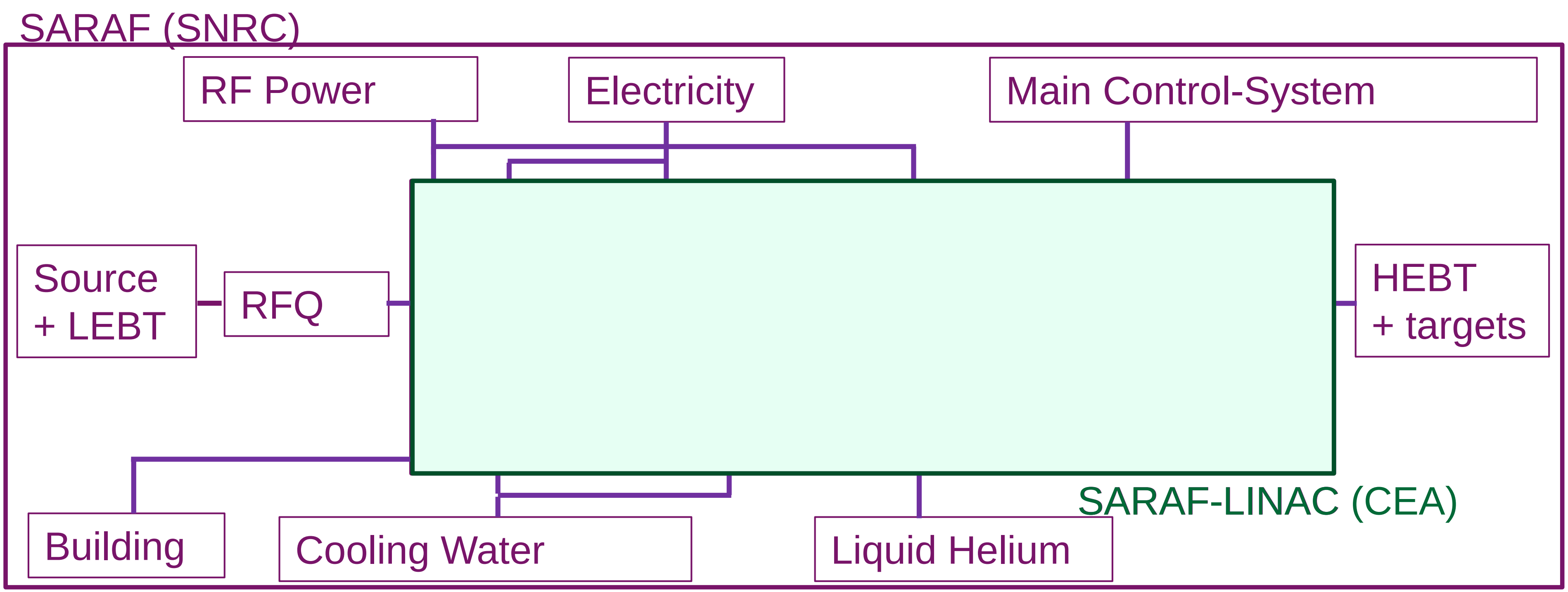

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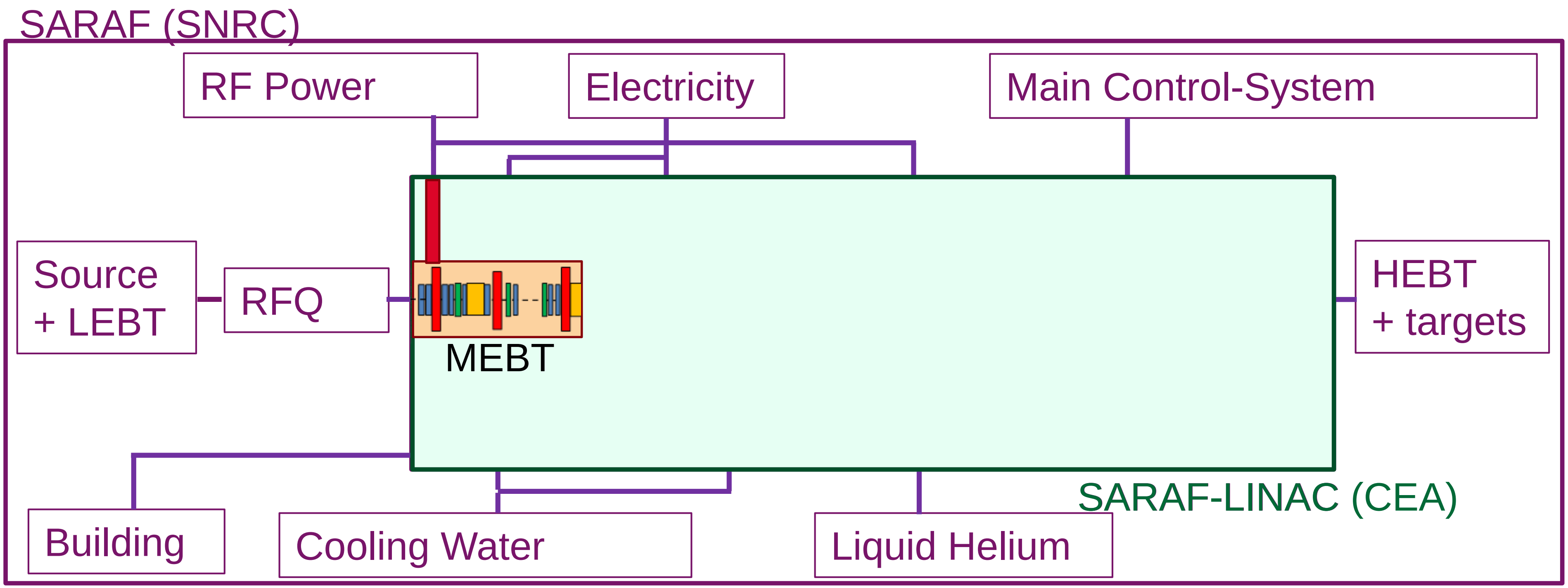

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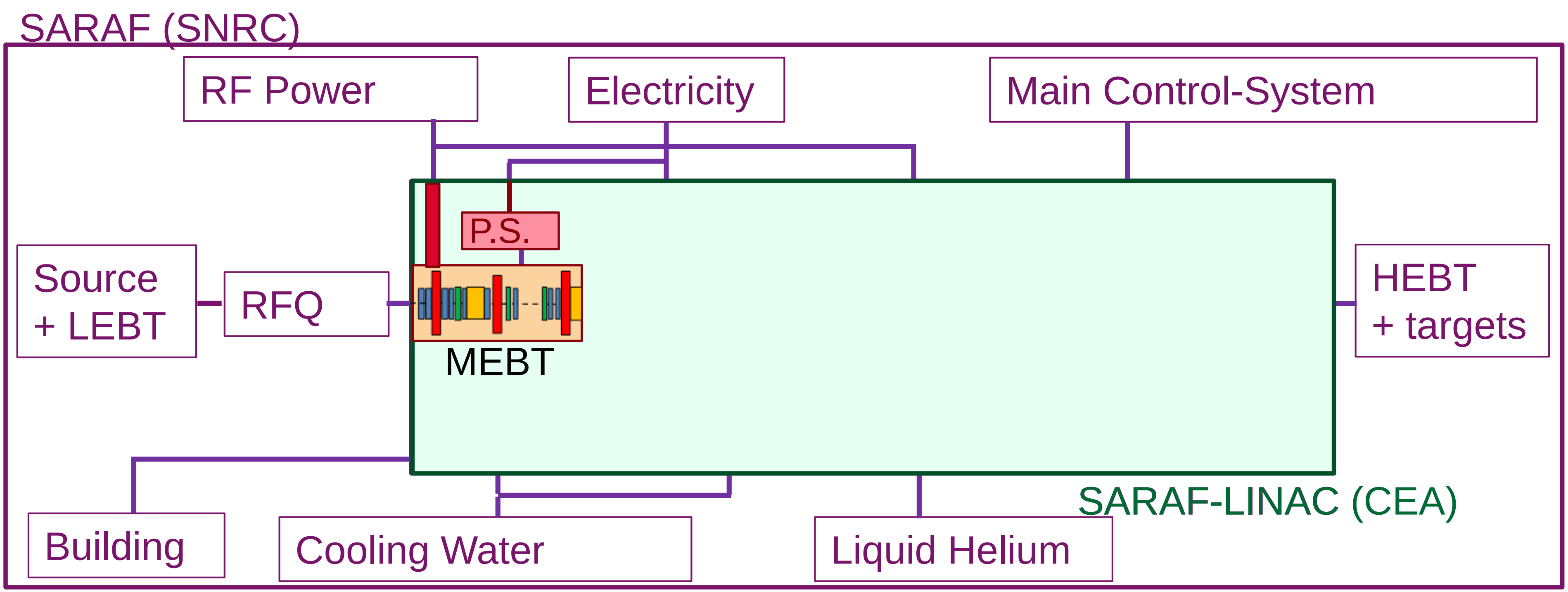

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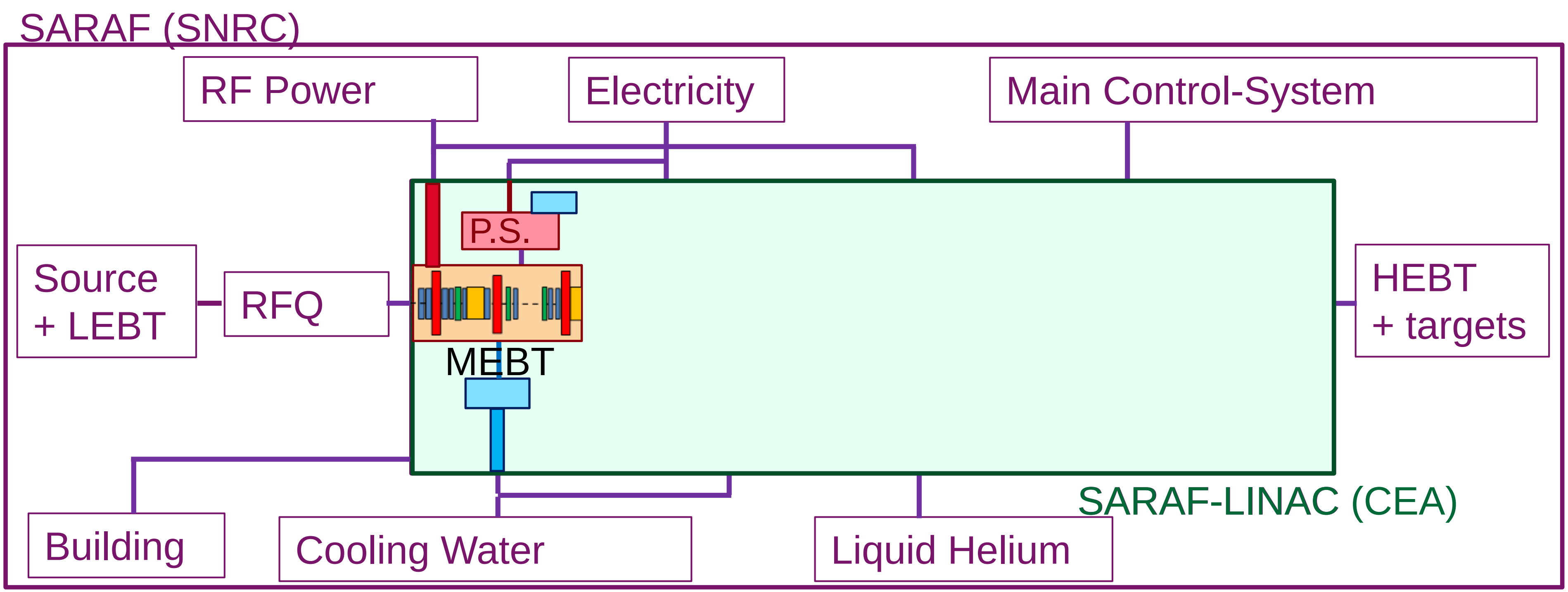

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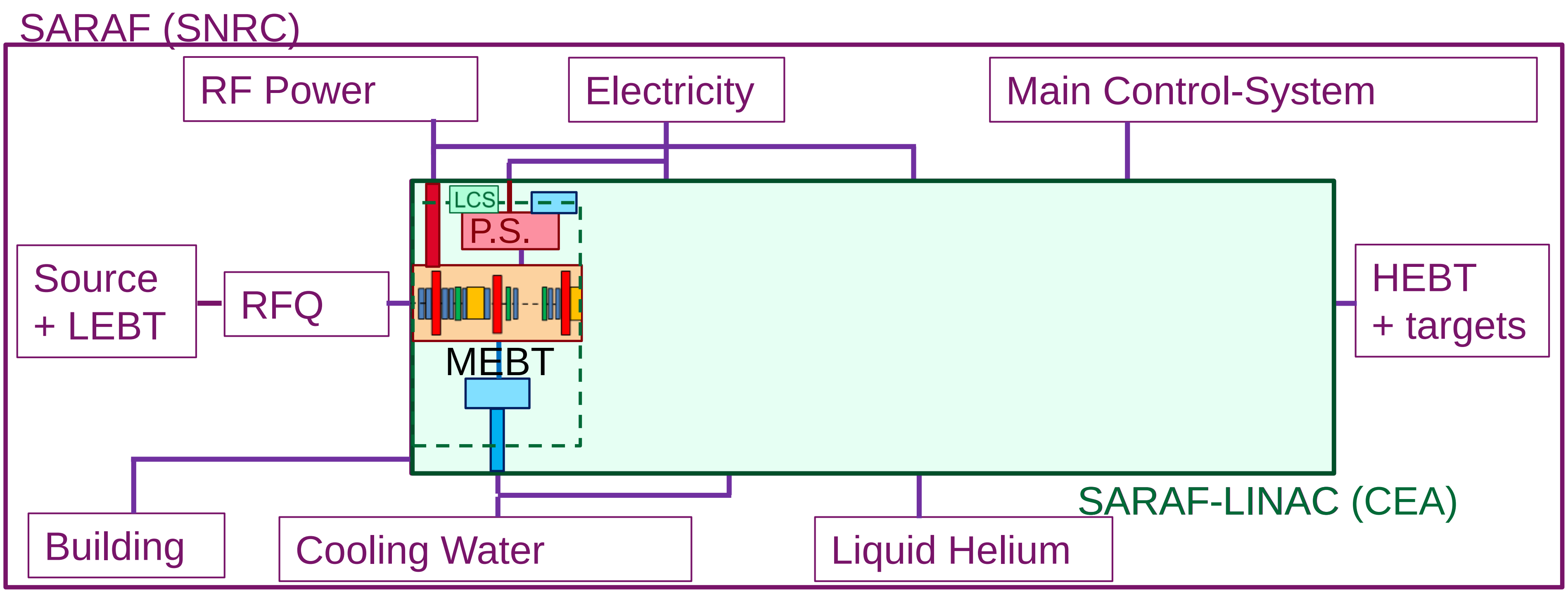

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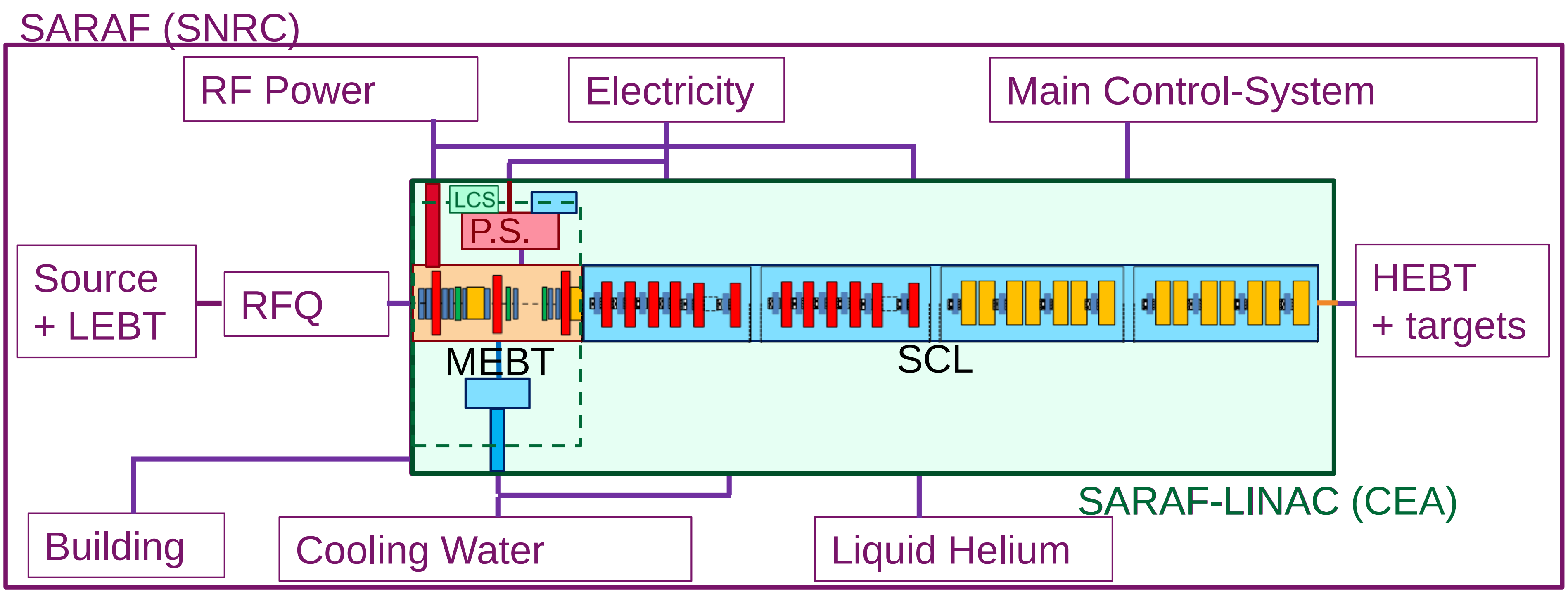

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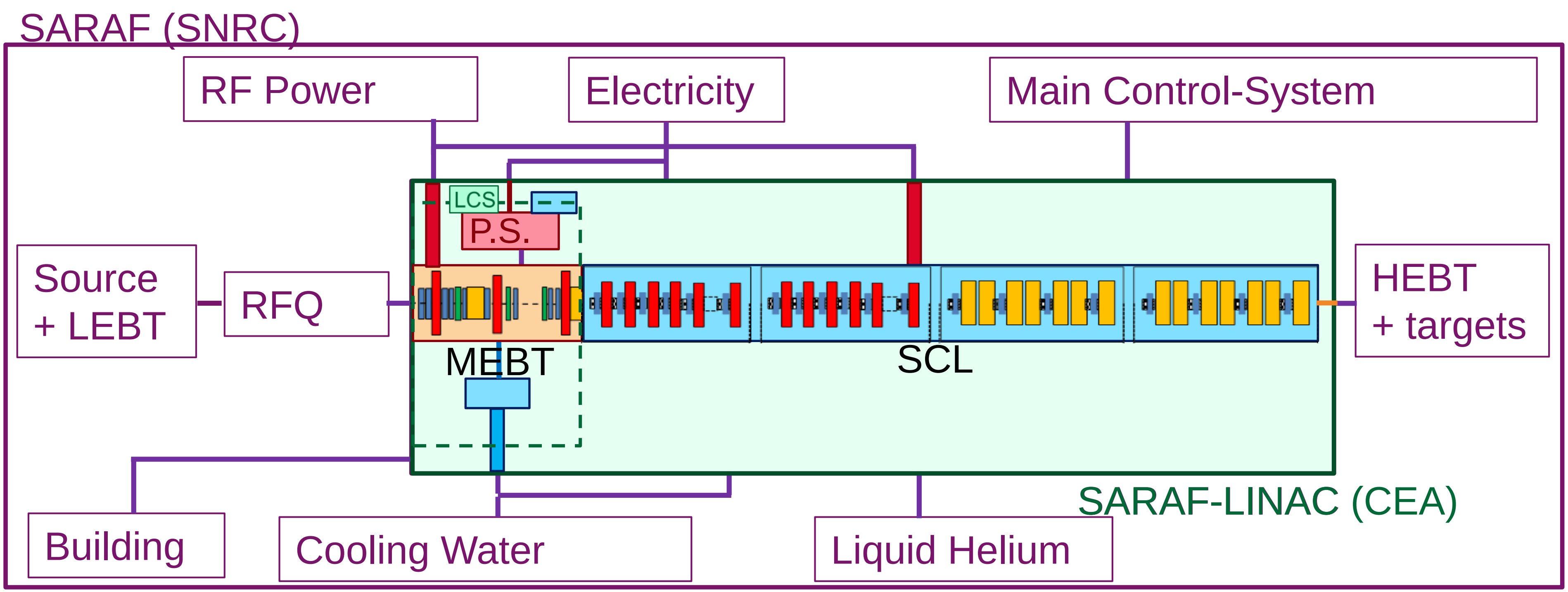

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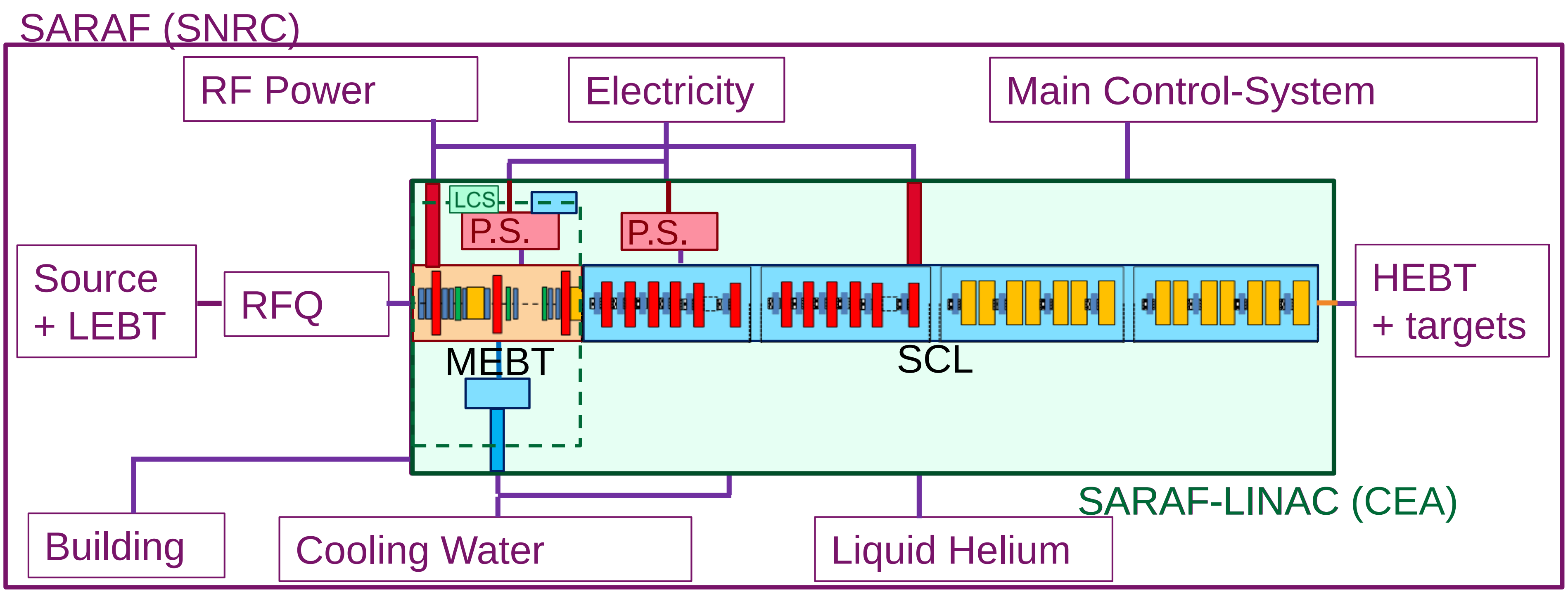

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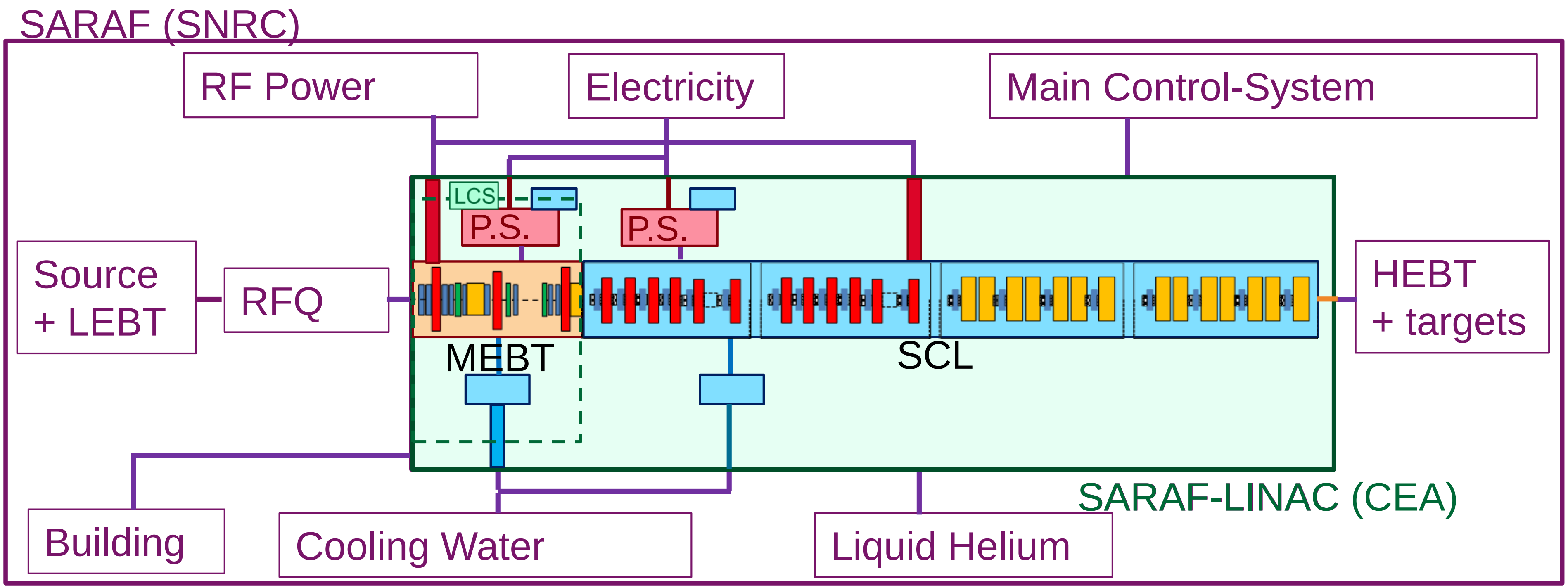

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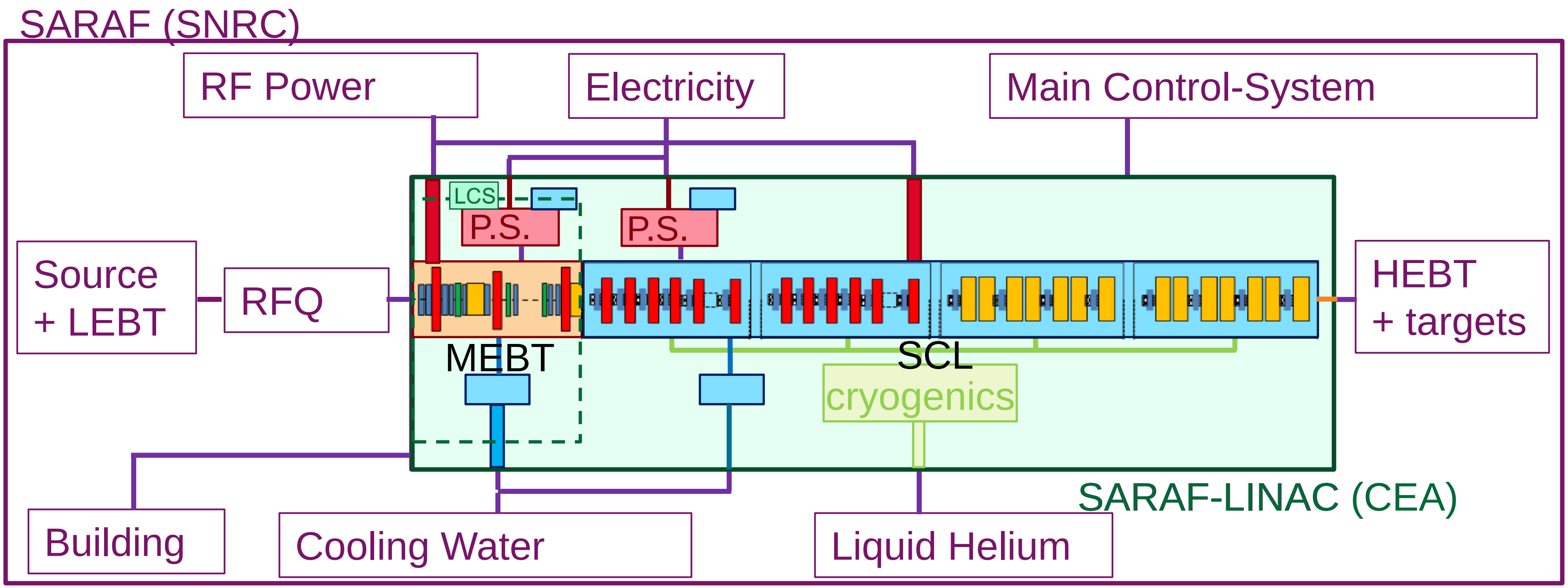

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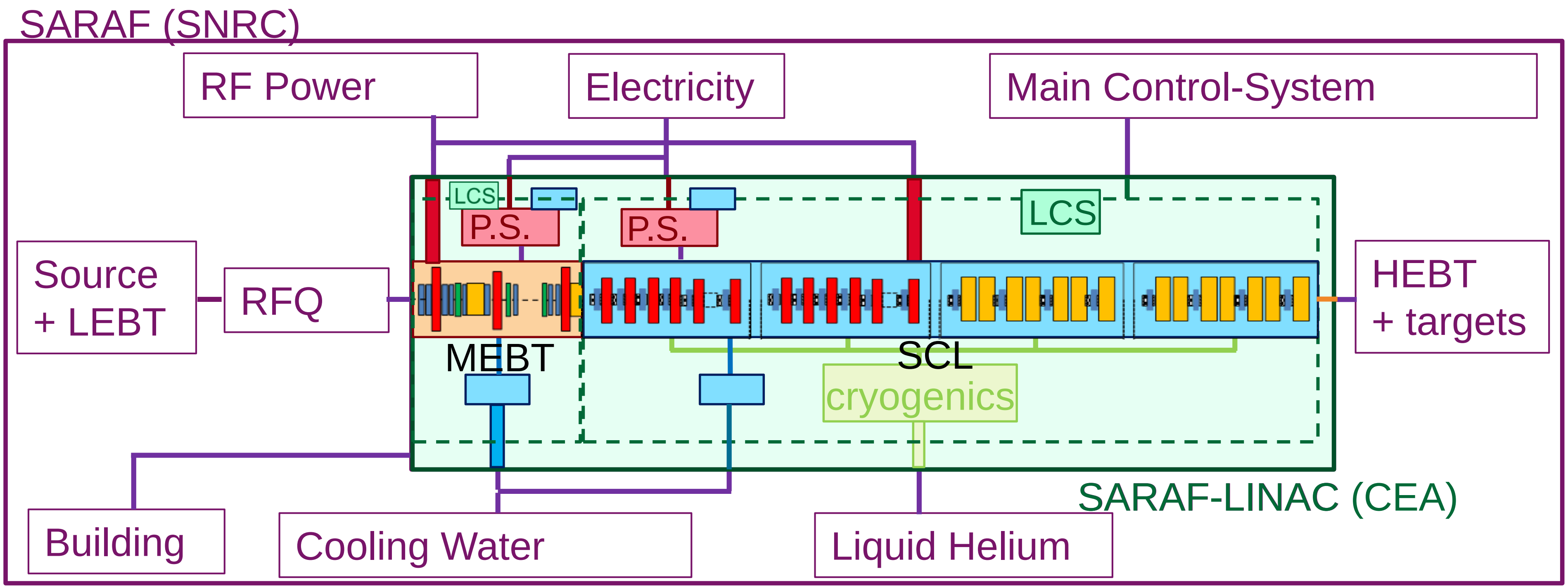
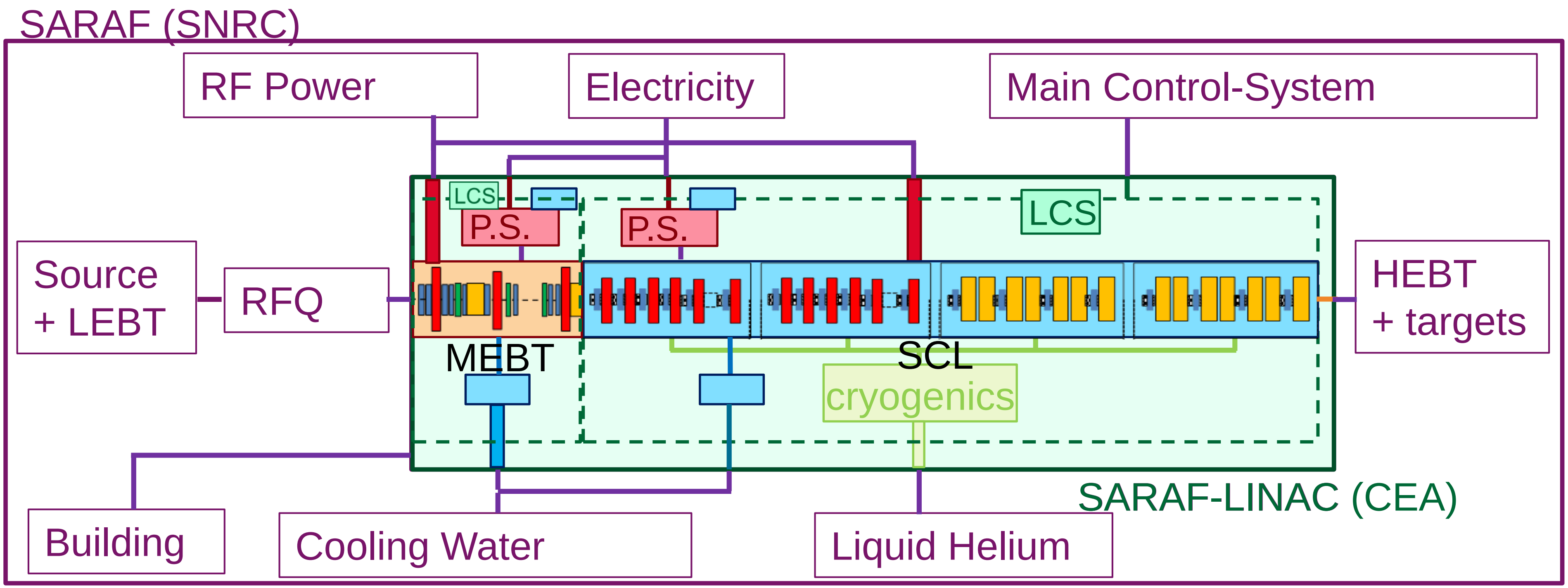


MAJOR COMPONENTS



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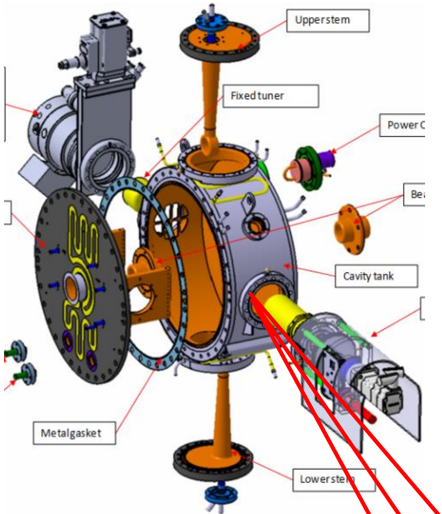

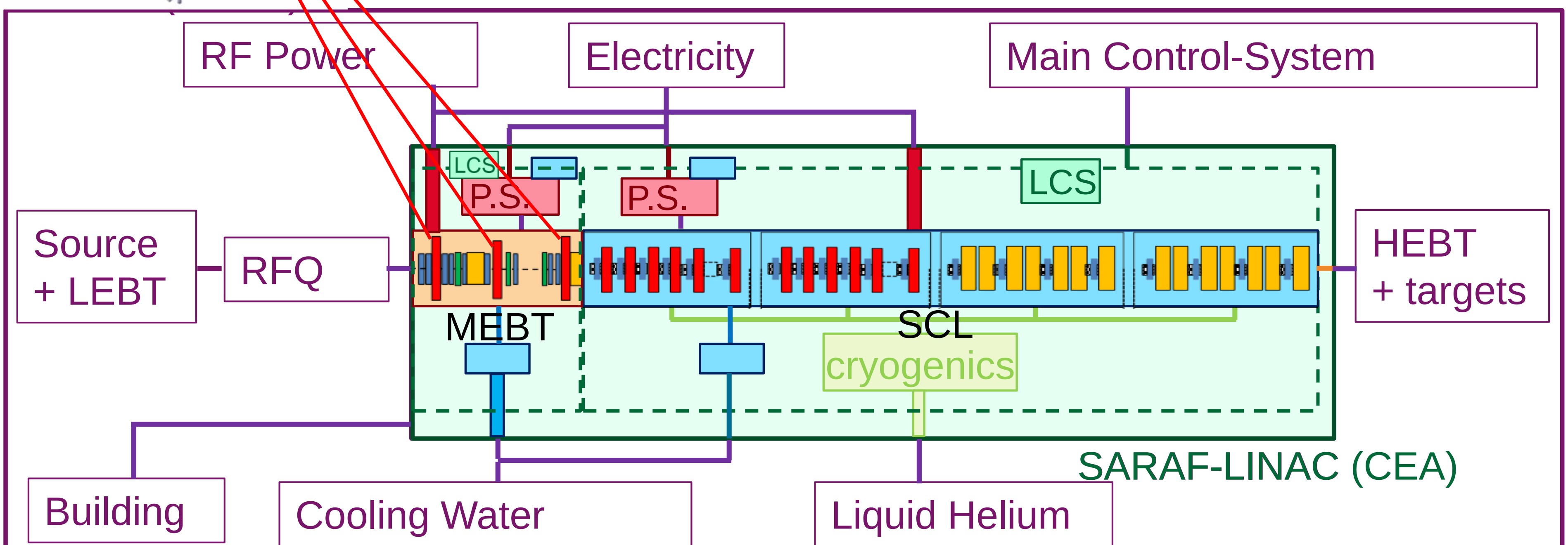

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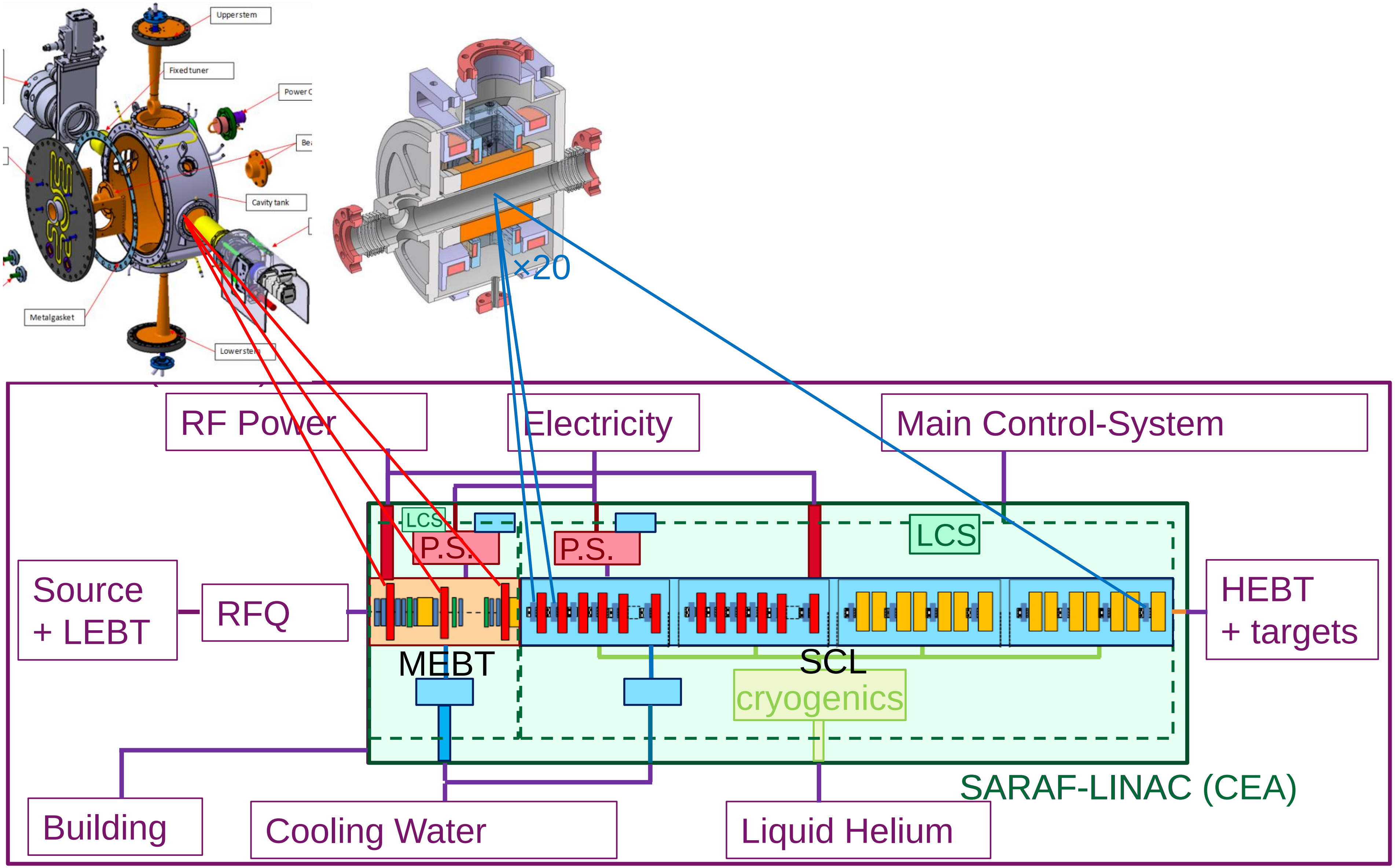

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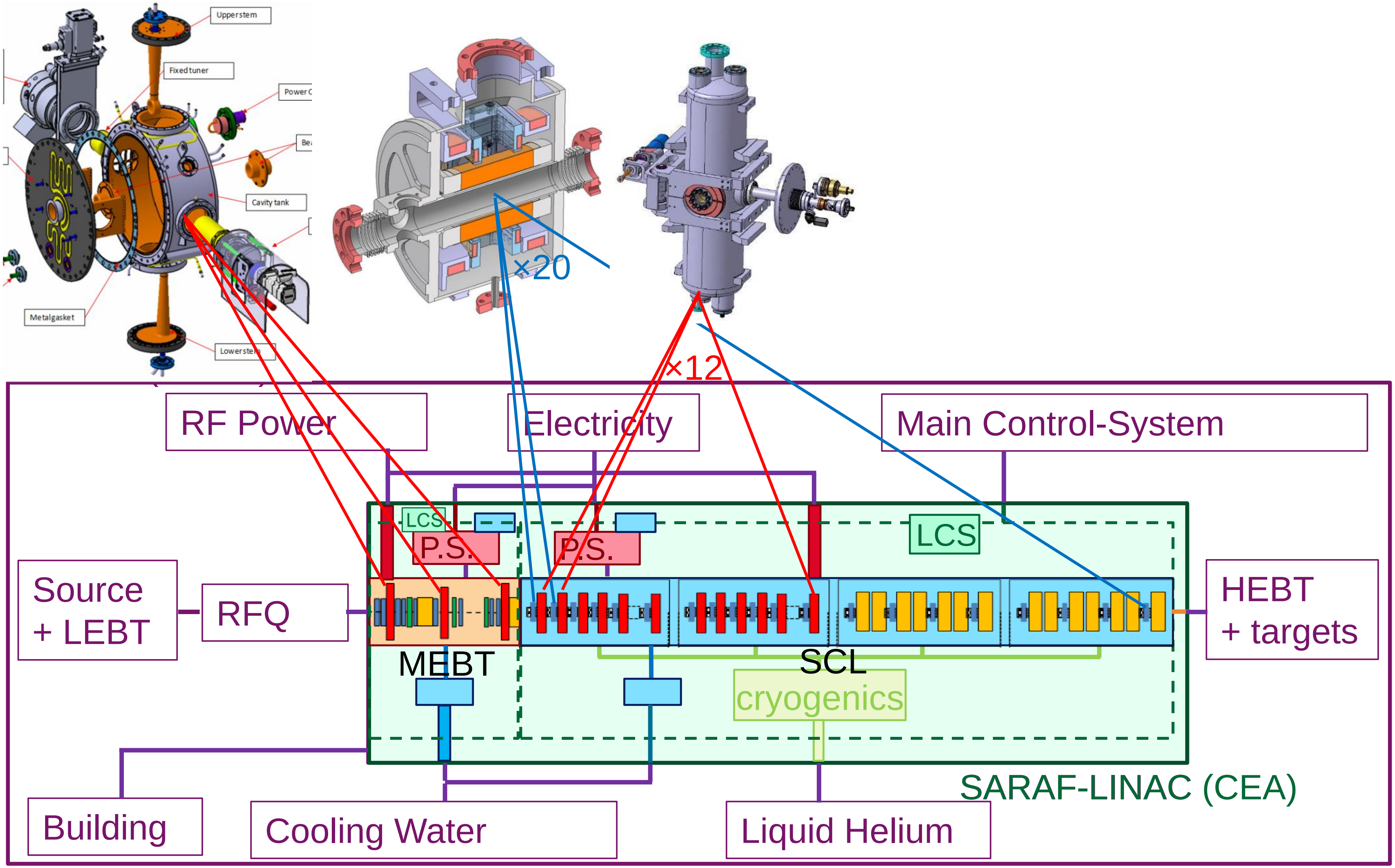

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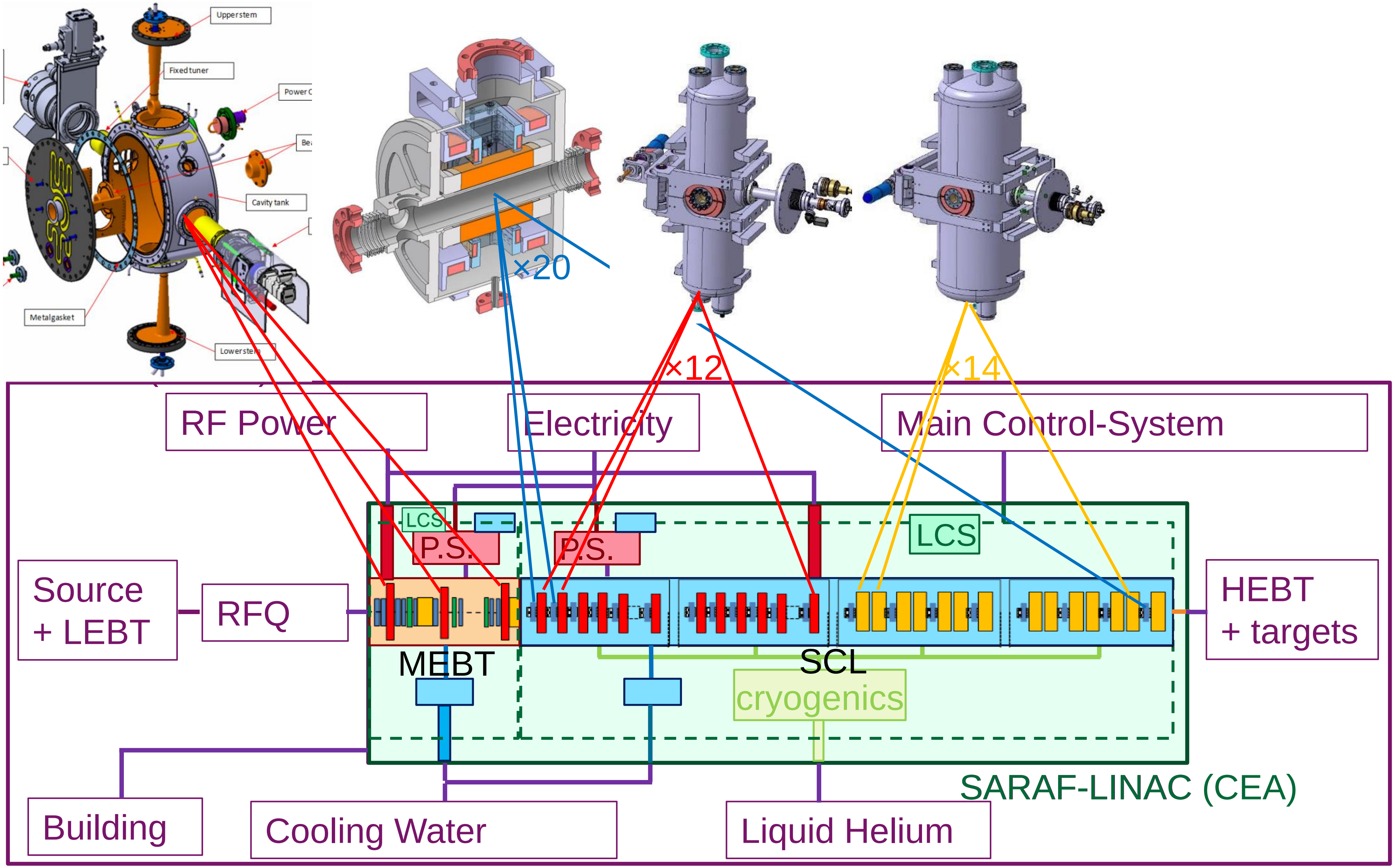

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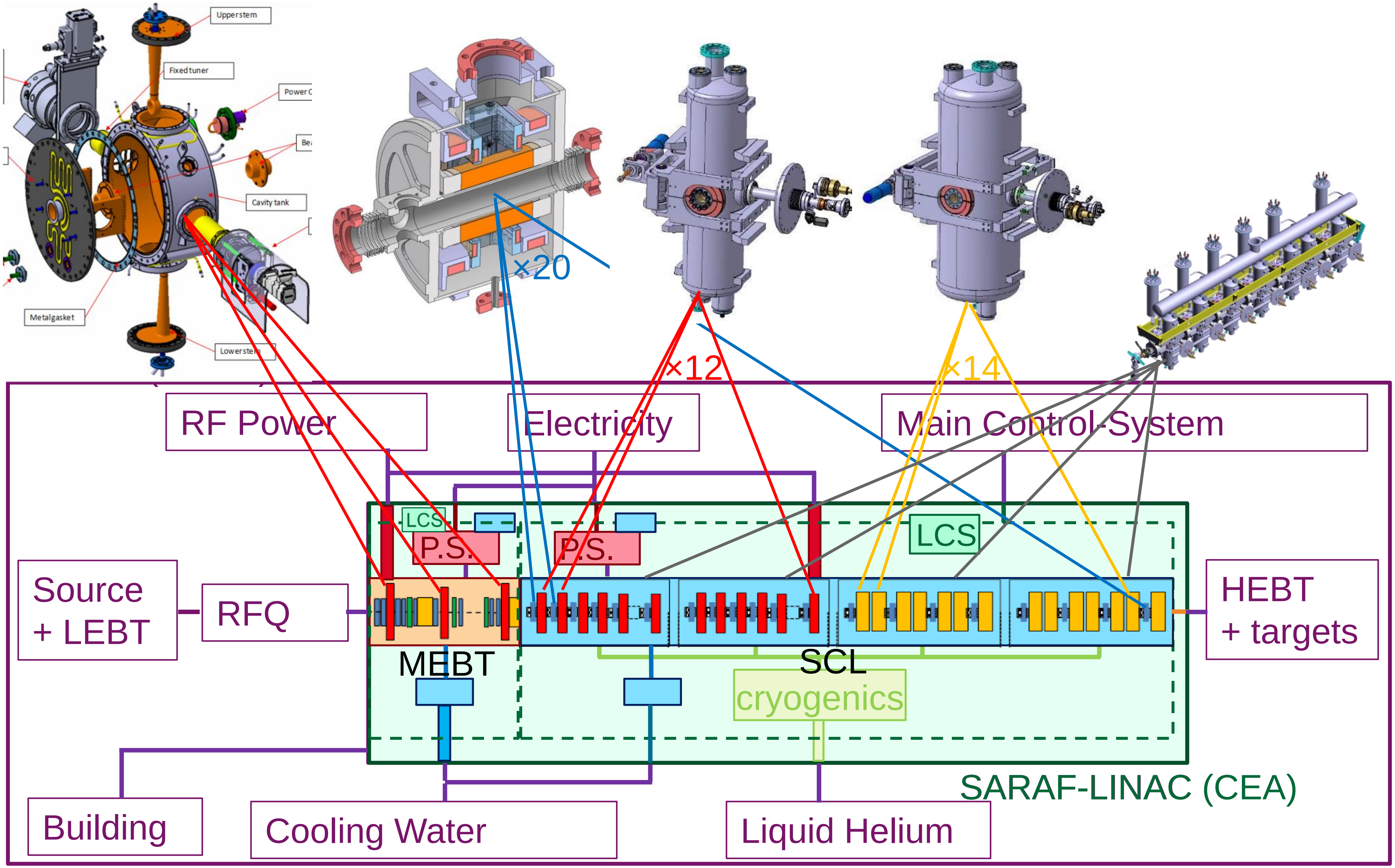

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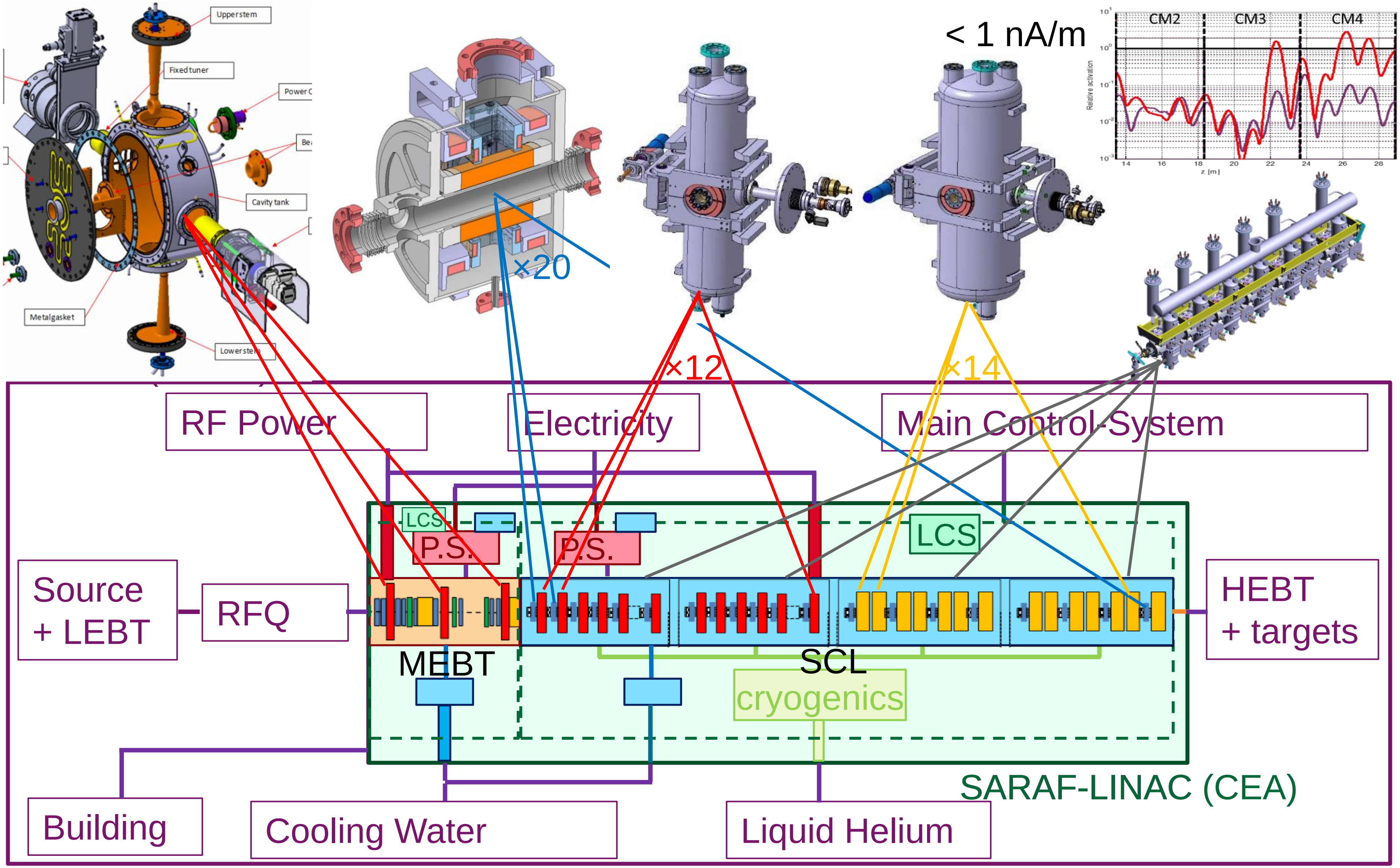
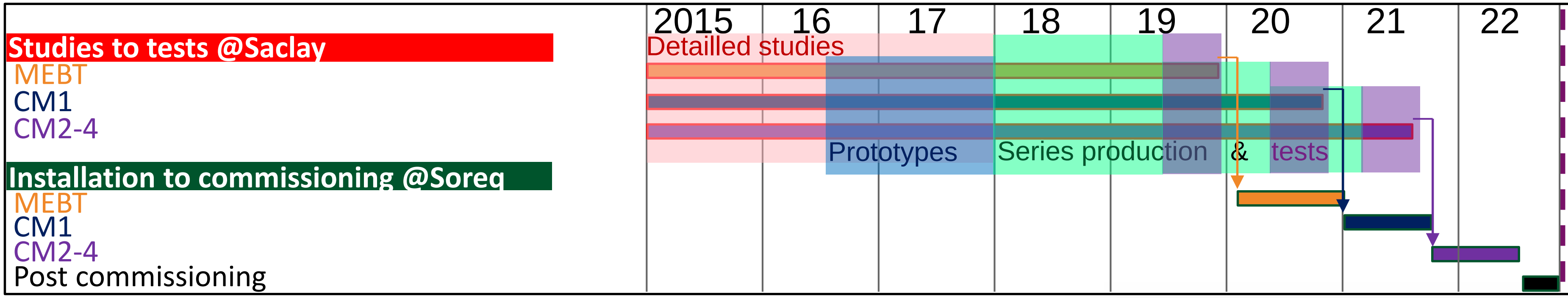


SARAF-LINAC PROJECT SCHEDULE



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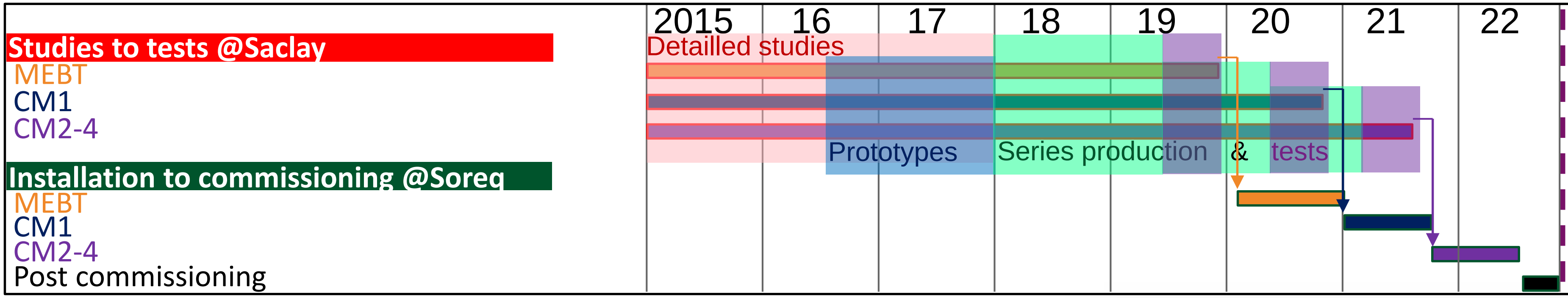

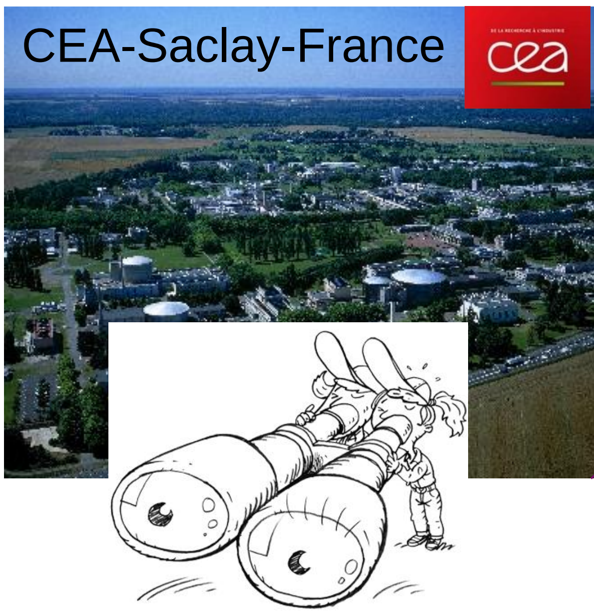

Detailed studies

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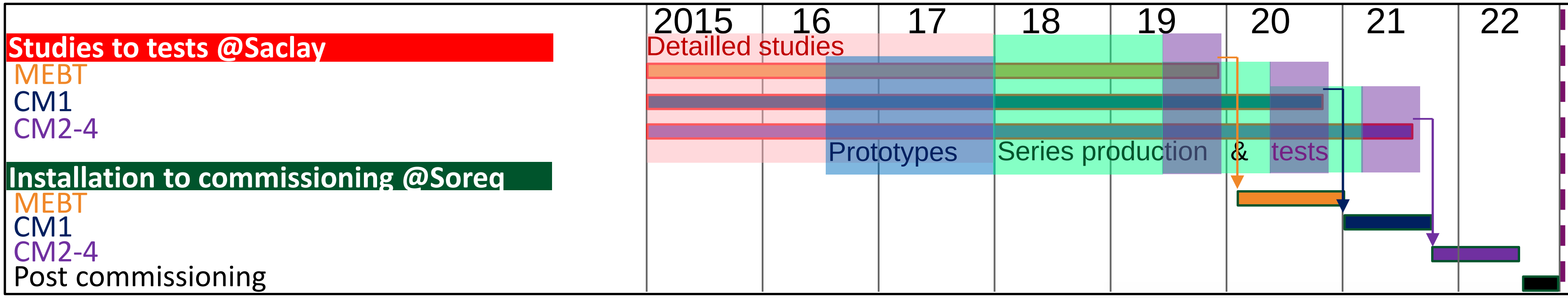

Call for tender

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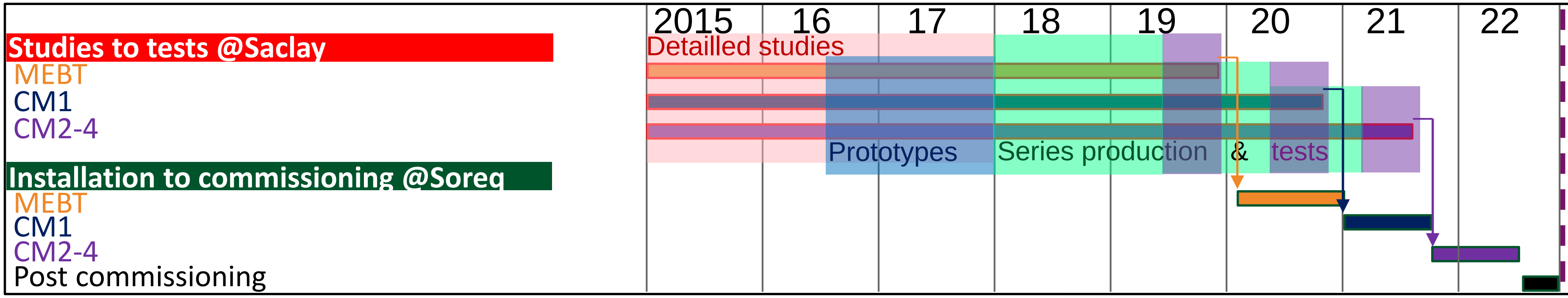

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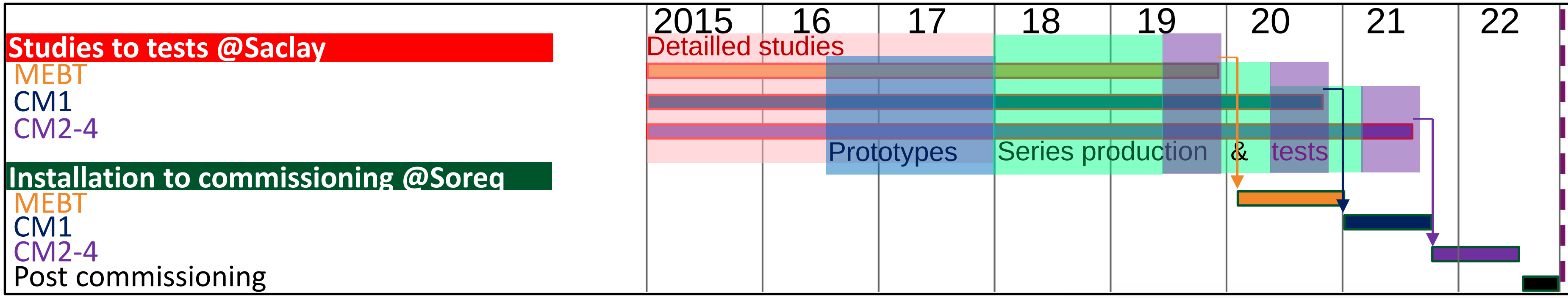

Prototyping

Series

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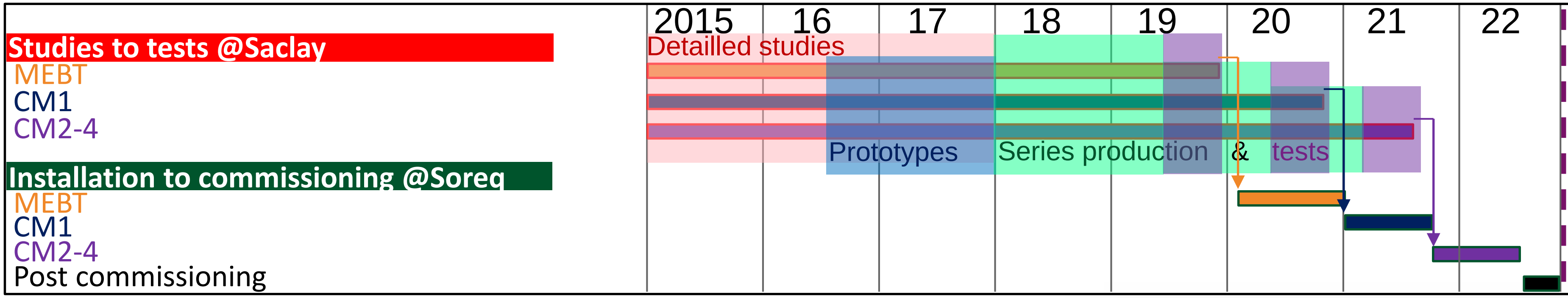

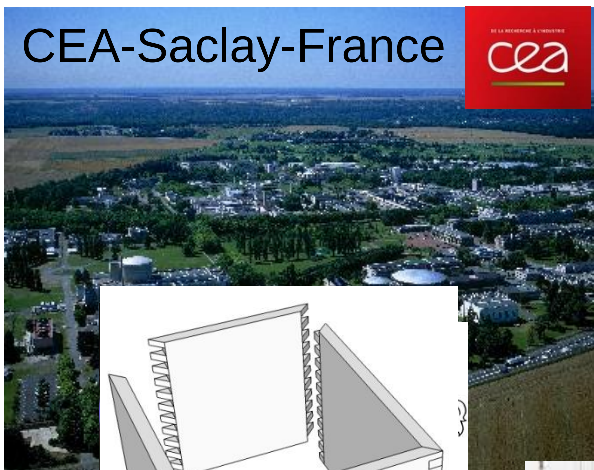

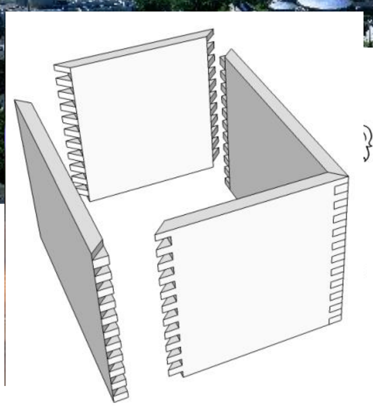

Test-Assemby

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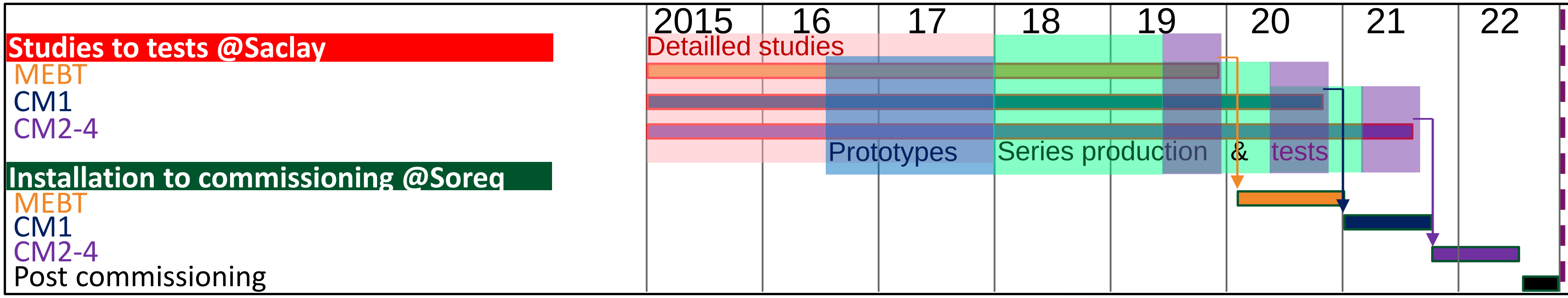

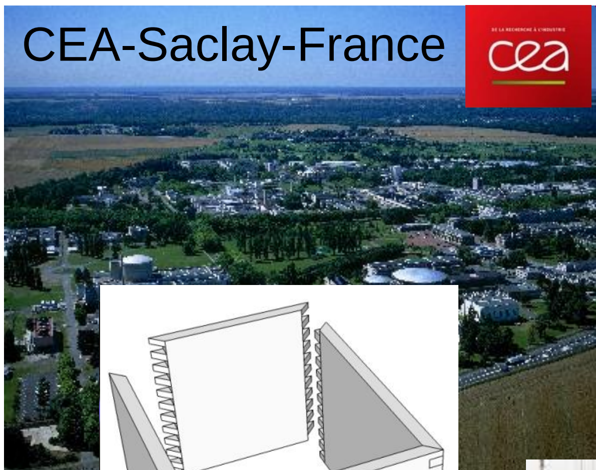

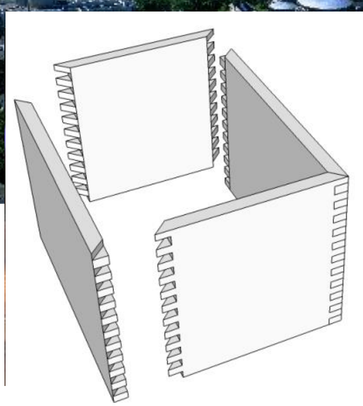

Test-Assemby

Series

Transport

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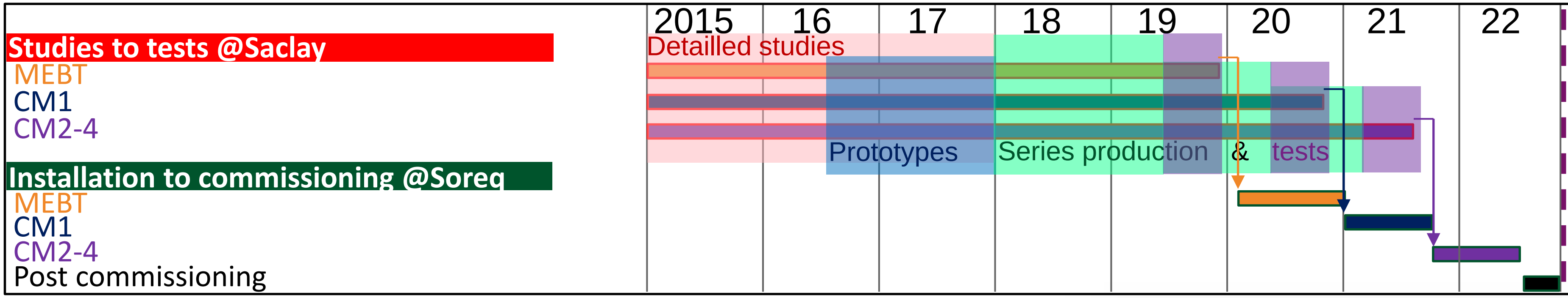

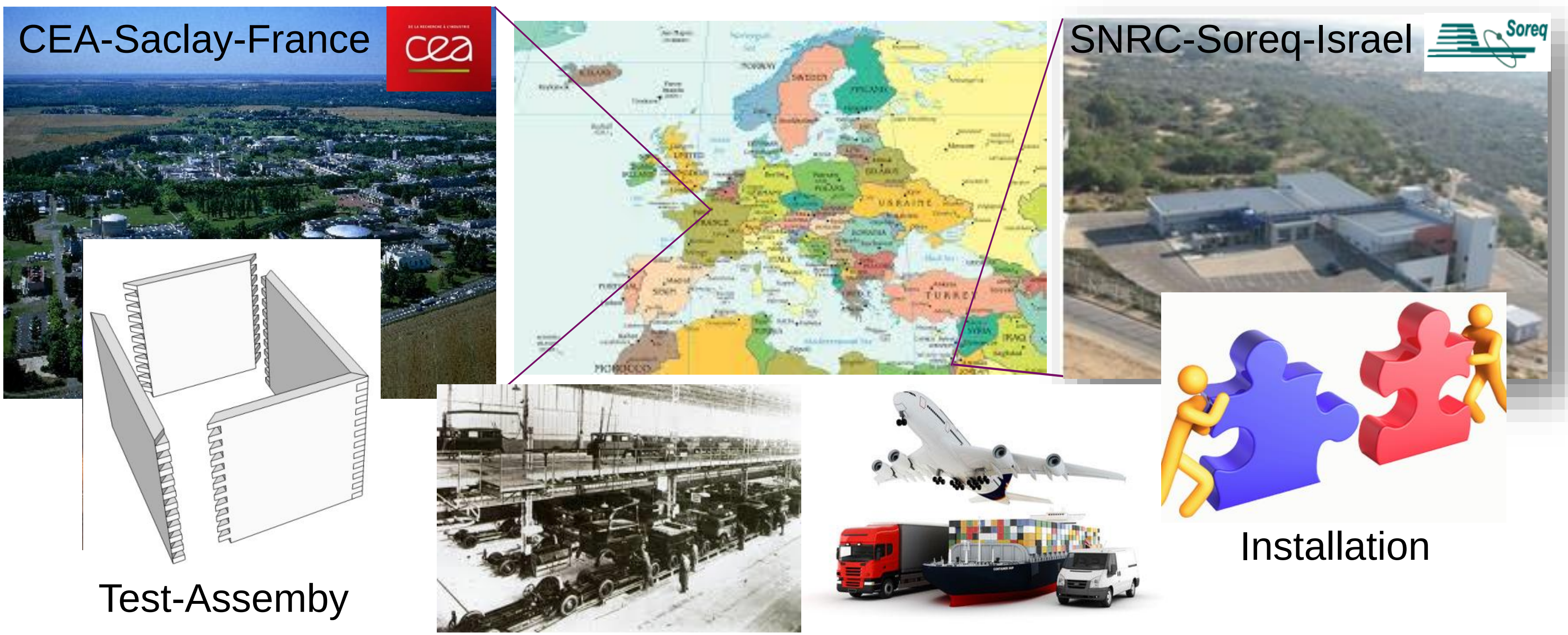

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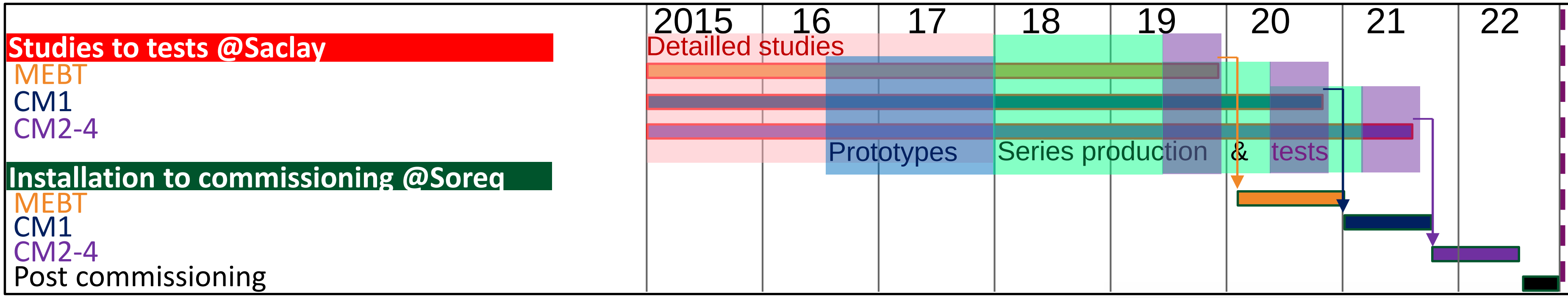

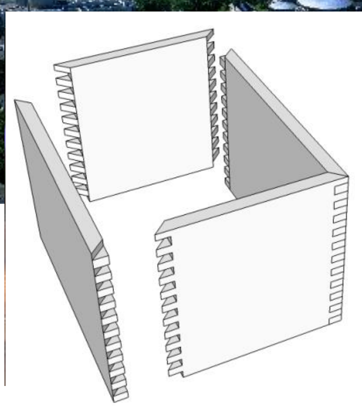

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commissioning.

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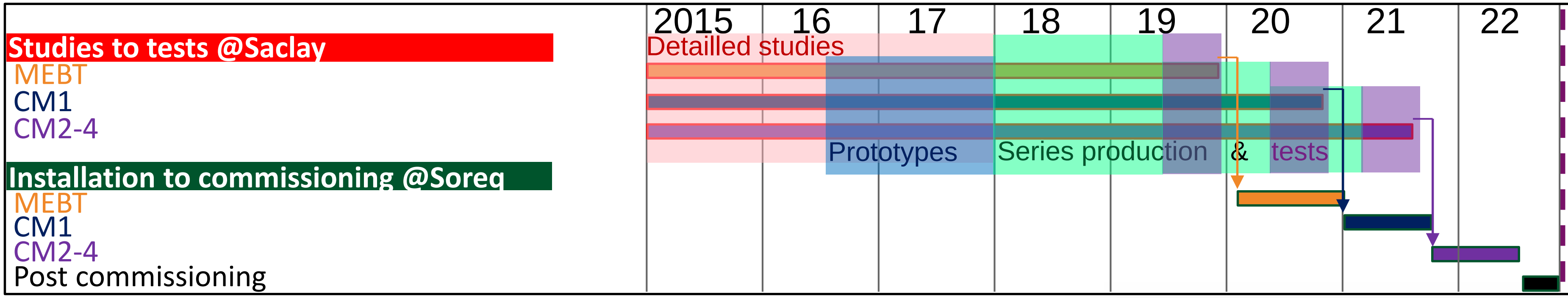

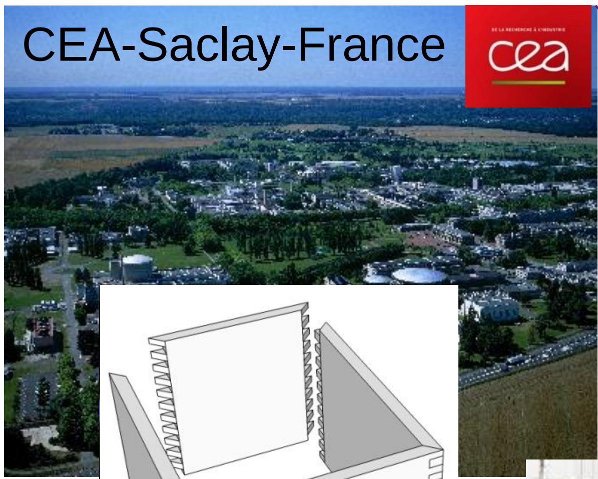

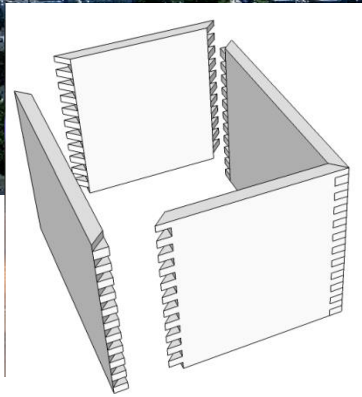

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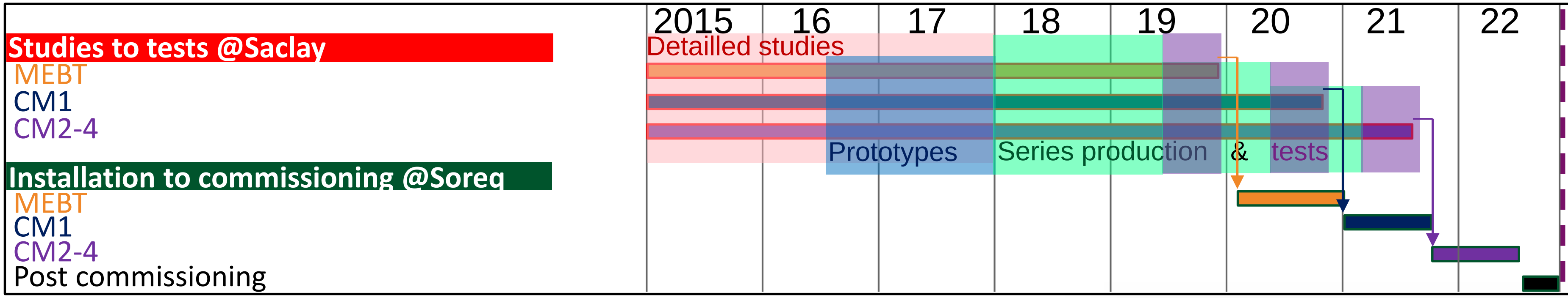

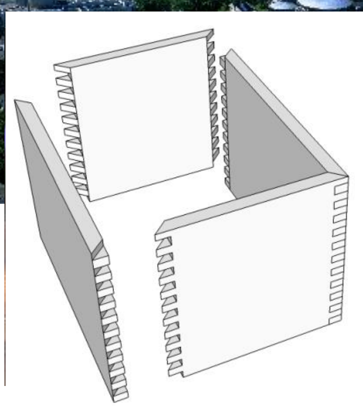

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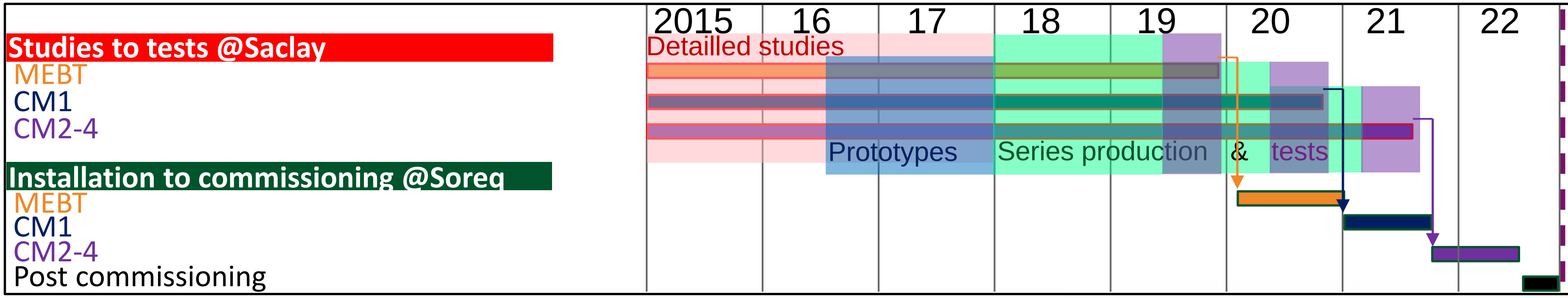
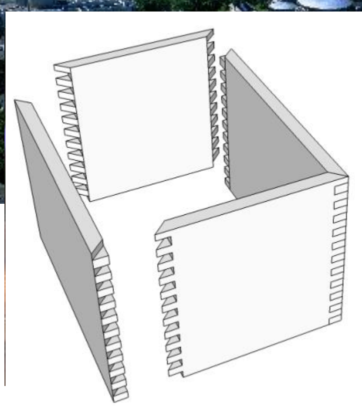



CEA-Saclay-France


SNRC-Soreq-Israel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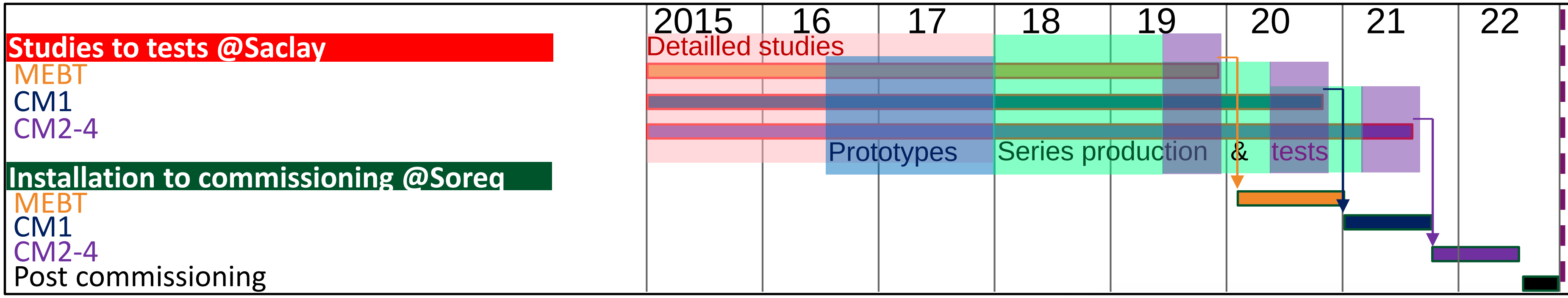

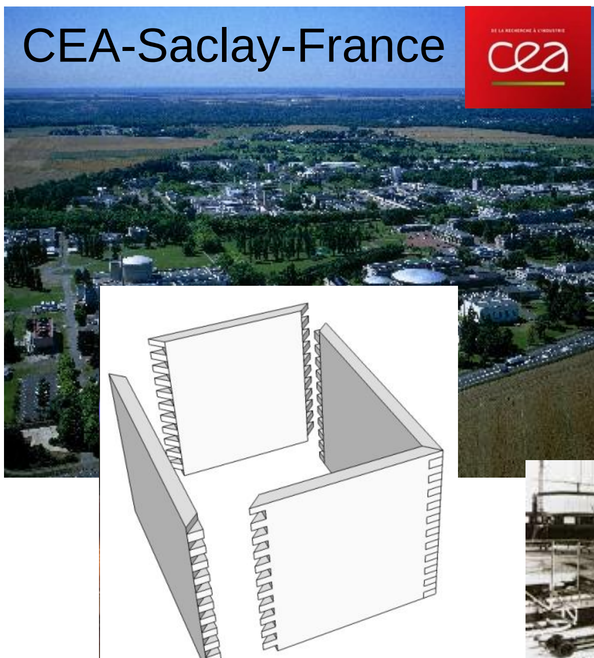

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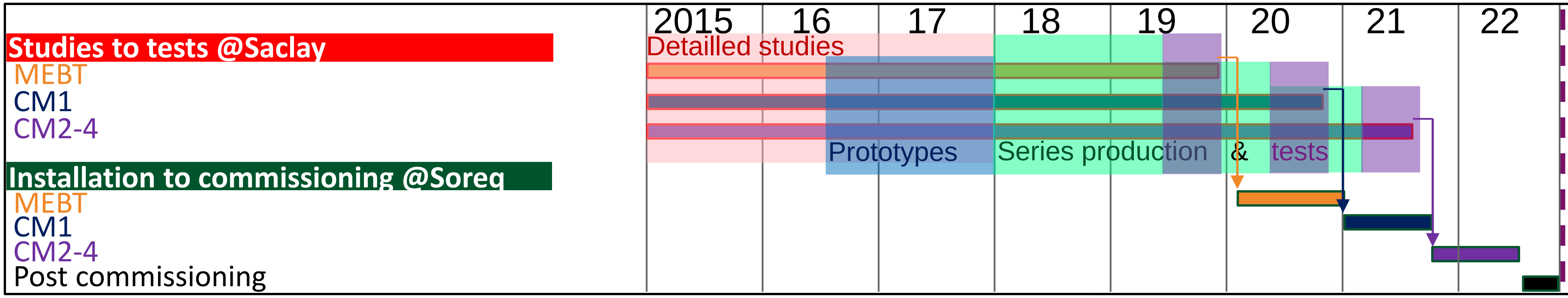

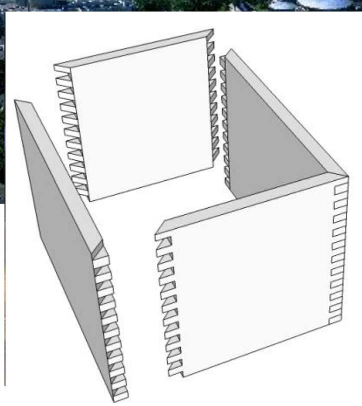

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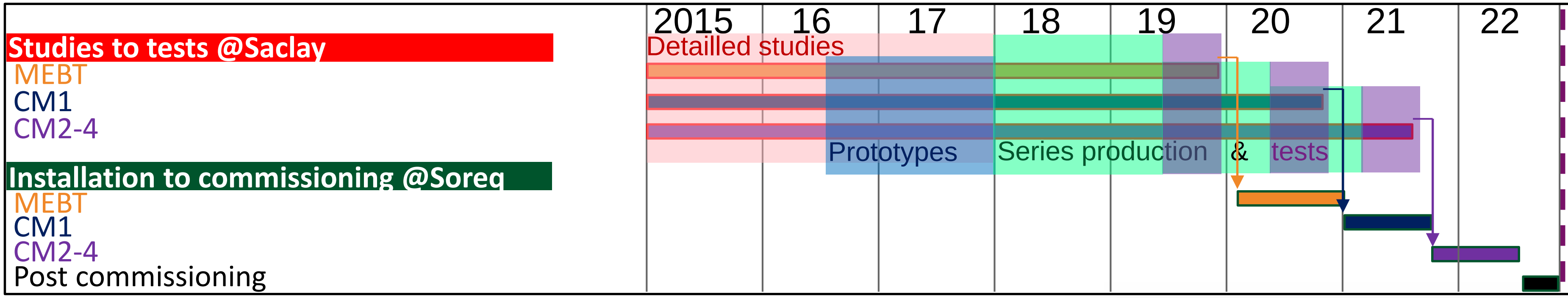

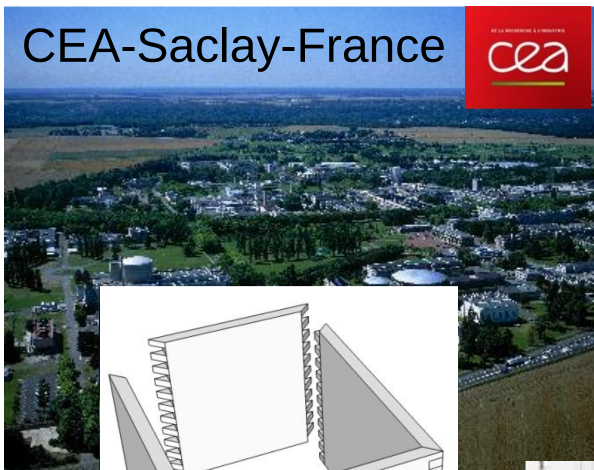

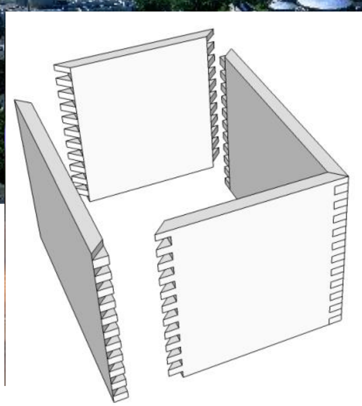

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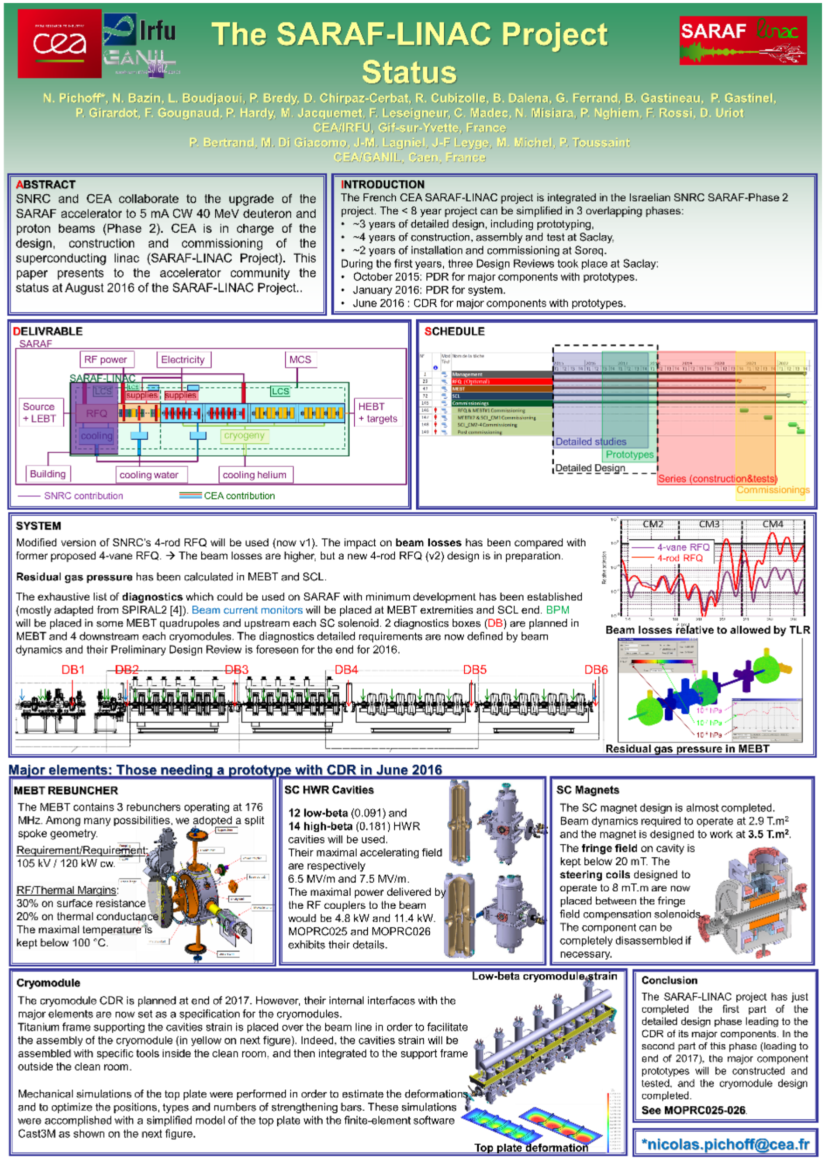
SARAF-LINAC POSTER

You are welcomed for discussions in front of the SARAF-LINAC posters

MOPRL075	This general description
MOPRL050	About MEBT rebuncher
MOPRC025 MOPRC026	About SC HWR cavities
TH1A04	About 4-rod RFQ



The SARAF-LINAC Project Status

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ABSTRACT
SNRC and CEA collaborate to the upgrade of the SARAF accelerator to 5 mA CW 40 MeV deuteron and proton beams (Phase 2). CEA is in charge of the design, construction and commissioning of the superconducting linac (SARAF-LINAC Project). This paper presents to the accelerator community the status at August 2016 of the SARAF-LINAC Project..

INTRODUCTION
The French CEA SARAF-LINAC project is integrated in the Israelian SNRC SARAF-Phase 2 project. The < 8 year project can be simplified in 3 overlapping phases:
- ~3 years of detailed design, including prototyping,
- ~4 years of construction, assembly and test at Saclay,
- ~2 years of installation and commissioning at Soreq.

During the first years, three Design Reviews took place at Saclay:
- October 2015: PDR for major components with prototypes.
- January 2016: PDR for system.
- June 2016 : CDR for major components with prototypes.

DELIVRABLE
SARAF, RF power, Electricity, MCS, SARAF-LINAC, LCS, supplies, supplies, LCS, Source + LEBT, RFQ, HEBT + targets, cooling, cryogeny, Building, cooling water, cooling helium
SNRC contribution — CEA contribution

SCHEDULE
Detailed studies, Prototypes, Detailed Design, Series (construction&tests), Commissionings

SYSTEM
Modified version of SNRC's 4-rod RFQ will be used (now v1). The impact on **beam losses** has been compared with former proposed 4-vane RFQ. → The beam losses are higher, but a new 4-rod RFQ (v2) design is in preparation.

Residual gas pressure has been calculated in MEBT and SCL.

The exhaustive list of **diagnostics** which could be used on SARAF with minimum development has been established (mostly adapted from SPIRAL2 [4]). Beam current monitors will be placed at MEBT extremities and SCL end. BPM will be placed in some MEBT quadrupoles and upstream each SC solenoid. 2 diagnostics boxes (DB) are planned in MEBT and 4 downstream each cryomodules. The diagnostics detailed requirements are now defined by beam dynamics and their Preliminary Design Review is foreseen for the end for 2016.

DB1 DB2 DB3 DB4 DB5 DB6

CM2 CM3 CM4 — 4-vane RFQ, 4-rod RFQ
Beam losses relative to allowed by TLR

Residual gas pressure in MEBT

Major elements: Those needing a prototype with CDR in June 2016

MEBT REBUNCHER
The MEBT contains 3 rebunchers operating at 176 MHz. Among many possibilities, we adopted a split spoke geometry.
Requirement/Requirement:
105 kV / 120 kW cw.
RF/Thermal Margins:
30% on surface resistance
20% on thermal conductance
The maximal temperature is kept below 100 °C.

SC HWR Cavities
12 low-beta (0.091) and **14 high-beta** (0.181) HWR cavities will be used.
Their maximal accelerating field are respectively 6.5 MV/m and 7.5 MV/m.
The maximal power delivered by the RF couplers to the beam would be 4.8 kW and 11.4 kW.
MOPRC025 and MOPRC026 exhibits their details.

SC Magnets
The SC magnet design is almost completed.
Beam dynamics required to operate at 2.9 $T.m^2$ and the magnet is designed to work at **3.5 $T.m^2$**.
The **fringe field** on cavity is kept below 20 mT. The **steering coils** designed to operate to 8 mT.m are now placed between the fringe field compensation solenoids.
The component can be completely disassembled if necessary.

Cryomodule
The cryomodule CDR is planned at end of 2017. However, their internal interfaces with the major elements are now set as a specification for the cryomodules.
Titanium frame supporting the cavities strain is placed over the beam line in order to facilitate the assembly of the cryomodule (in yellow on next figure). Indeed, the cavities strain will be assembled with specific tools inside the clean room, and then integrated to the support frame outside the clean room.

Mechanical simulations of the top plate were performed in order to estimate the deformations and to optimize the positions, types and numbers of strengthening bars. These simulations were accomplished with a simplified model of the top plate with the finite-element software Cast3M as shown on the next figure.

Low-beta cryomodule strain

Top plate deformation

Conclusion
The SARAF-LINAC project has just completed the first part of the detailed design phase leading to the CDR of its major components. In the second part of this phase (leading to end of 2017), the major component prototypes will be constructed and tested, and the cryomodule design completed.
See MOPRC025-026.

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